# PROJECT REPORT

OBJECT-ORIENTED PROGRAMMING
(OOP)

GYM MEMBERSHIP MANAGEMENT SYSTEM

# GROUP MEMBERS

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# 1. INTRODUCTION

The Gym Membership Management System is a console-based application developed in C++ using the principles of Object-Oriented Programming (OOP). The purpose of this project is to automate the management of gym members by replacing manual record-keeping with a digital solution.
The application allows users to add, search, update, display, and delete member records while storing data permanently using text files. It also implements exception handling to improve reliability and prevent unexpected program crashes.

# 2. PROJECT OBJECTIVES

- Develop a practical application using C++ and OOP concepts.
- Simplify gym membership management.
- Store member information permanently using file handling.
- Validate user input and handle runtime errors safely.
- Demonstrate abstraction, inheritance, polymorphism, and encapsulation in a real-world project.

# 3. ORGANIZATION AND STRUCTURE

The project follows a modular and object-oriented design to improve readability and maintainability.

## MAIN CLASSES

### PERSON (ABSTRACT BASE CLASS)

- Stores common member information.
- Provides pure virtual functions.
- Demonstrates abstraction.

### MEMBER (DERIVED CLASS)

- Inherits from the Person class.
- Stores email, phone number, membership type, attendance, fee, and status.
- Implements overridden display and file functions.

### FILE MANAGER

- Handles reading and writing data.
- Saves records into members.txt.
- Loads saved records when the program starts.

### GYM SYSTEM

- Controls the complete application.
- Displays menus.
- Performs add, update, search, delete, and display operations.

The separation of responsibilities makes the system organized and easier to extend.

## 4. RESEARCH AND INFORMATION GATHERING

Before developing the project, research was conducted on:

- Fundamentals of Object-Oriented Programming in C++.
- File handling using ifstream and ofstream.
- Exception handling using try and catch blocks.
- Input validation techniques for email addresses and phone numbers.
- Basic requirements of gym membership management systems.

Information was gathered from C++ programming books, online documentation, educational resources, and classroom lectures to ensure that the implementation followed standard programming practices.

## 5. PROCEDURE

The development process followed these steps:

1. Planned the project requirements and identified system features.
2. Designed the class hierarchy and relationships.
3. Implemented the abstract Person class.
4. Created the Member class through inheritance.
5. Added menu-driven functionality for user interaction.
6. Implemented file handling for permanent data storage.
7. Added exception handling and input validation.
8. Tested all functions including add, search, update, delete, and display.
9. Fixed errors and optimized the final program.

## 6. RESULTS AND DISCUSSION

The project was successfully completed and all major functionalities worked as expected.

### ACHIEVED RESULTS

- Members can be added successfully.
- Existing records can be searched using Member ID.
- Information can be updated without affecting other records.
- Members can be deleted safely.
- Data remains stored after closing the application.
- Saved records are automatically loaded on startup.
- Invalid input is handled through exception handling.

The implementation demonstrates that OOP principles improve code organization, reusability, and maintainability while file handling ensures data persistence.

# 7. OOP CONCEPTS IMPLEMENTED

The project successfully demonstrates multiple Object-Oriented Programming concepts.

### CLASSES AND OBJECTS

Different classes such as Person, Member, FileManager, and GymSystem organize the application into logical components.

### ENCAPSULATION

Data members are declared private and accessed through public getter and setter functions.

### INHERITANCE

The Member class inherits common properties and functions from the Person base class.

### POLYMORPHISM

Virtual functions are overridden in the derived class to provide specialized behavior.

### ABSTRACTION

The Person class contains pure virtual functions, making it an abstract base class.

### STATIC MEMBERS

Static variables keep track of the total number of objects.

### FRIEND FUNCTION

A friend function is used to access private information for displaying summaries.

### CONST FUNCTIONS

Getter functions are declared as const to ensure data is not modified unintentionally.

# 8. FILE HANDLING AND EXCEPTION HANDLING

## FILE HANDLING

The project uses:

- ofstream to save member records.
- ifstream to load saved records.
- Automatic loading of data during program startup.
- Permanent storage in members.txt.

## EXCEPTION HANDLING

The project uses try-catch blocks to handle:

- Invalid numeric input.
- Incorrect Member IDs.
- Missing data files.
- Runtime file access errors.

These techniques make the application more reliable and user-friendly.

# 9. CONCLUSION

The Gym Membership Management System successfully fulfills the objectives of the project by combining Object-Oriented Programming, file handling, and exception handling into a practical application.

The project demonstrates strong software design principles while providing an efficient solution for managing gym membership records. It serves as an excellent example of applying theoretical OOP concepts to solve real-world problems.

## 10. FUTURE IMPROVEMENTS

- Add administrator login authentication.
- Store data in a database instead of text files.
- Generate monthly fee and attendance reports.
- Add payment tracking functionality.
- Develop a graphical user interface (GUI).
- Increase the maximum member capacity dynamically.